

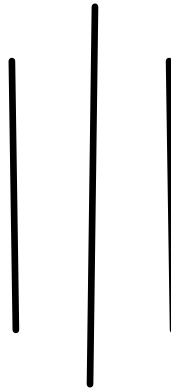
Tribhuvan University
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Prithvi Narayan Campus
BSc. CSIT Program

Case Study Report

**An Integrated E-Governance Platform for Digital Public Services
in Nepal**



Submitted To

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Abstract

Integrated e-governance is an approach to delivering government services through interconnected digital platforms. This study examines Nepal's efforts toward integrated digital public service delivery, using the Nagarik App as a primary example. Nepal has introduced various online government services, but many services and information systems have historically operated across separate platforms. The Nagarik App was developed as a service delivery gateway through which citizens can access online services from different government and public agencies [1]. The Government of Nepal has also continued to expand its digital public-service infrastructure through initiatives aimed at improving service accessibility and integration.

The analysis identifies both opportunities and challenges in Nepal's transition toward integrated e-governance. Integration can reduce the need for citizens to interact with multiple service platforms and can improve the accessibility and efficiency of public services. However, challenges remain in areas such as interoperability, fragmented government information systems, digital infrastructure, cybersecurity, data governance, and unequal access to digital services [2]. These challenges are particularly important as more government services become dependent on interconnected digital systems.

Nepal's Digital Transformation Project, approved by the World Bank in 2026, further demonstrates the country's move toward integrated digital public infrastructure. The project includes an integrated online citizen service portal, a secure government-wide data exchange, and a digital locker for managing verifiable credentials and digital documents [2]. Based on the case analysis, this study recommends strengthening interoperability through secure data-exchange mechanisms and common standards, improving nationwide digital connectivity and digital literacy, enhancing coordination among government agencies, and establishing stronger data-governance and cybersecurity practices. A conceptual architecture and phased approach for integrating digital identity, authentication, data exchange, and public services are also proposed. Overall, Nepal has made significant progress toward integrated digital public service delivery, but sustained technical, institutional, and policy improvements are required to achieve a more connected, secure, and inclusive e-governance ecosystem.

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Chapter 1: Introduction

1. Background of the Study

E-Governance is the use of information and communication technologies (ICT) to improve government service delivery, transparency and accountability [2]. Unlike traditional e-government (which may simply move forms online), integrated e-governance emphasizes one-stop digital access: a unified platform where citizens can authenticate once and access multiple government services. Such integration can reduce duplication, improve interoperability between agencies, and place the citizen at the center of public services. In ideal terms, an integrated e-governance platform allows citizens to use a single digital identity to interact with various government databases (tax, health, legal documents, etc.) securely and conveniently [1][6].

Globally, integrated platforms have become central to digital-government strategies. For example, the World Bank's press release (February 2026) highlights Nepal's new Digital Transformation Project, which "will invest in digital platforms including an integrated online citizen service portal and an integrated social registry... [and] a secure government-wide data exchange" [1]. This mirrors similar approaches in other countries: India's "One Nation, One Portal" (UMANG) or Estonia's X-Road infrastructure. In summary, integrated e-governance aims to make public services more accessible, efficient and transparent by breaking down silos between government systems.

2. E-Governance in Nepal

Nepal has been actively pursuing digital governance over the past two decades. Foundational policies include the E-Government Master Plan (2006) and the Digital Nepal Framework (2019) [3]. These set targets for ICT adoption across sectors. In practice, Nepal has deployed many e-services: online tax (PAN) filing, e-procurement (e-GP), electronic licensing (driving, vehicle), civil registration (birth/death/marriage), and local municipal portals. The Nagarik App (launched 2021) exemplifies Nepal's push for integration – it was explicitly conceived as a unified citizen-services platform to simplify access and digital identity [3].

Despite these initiatives, progress is mixed. A recent study notes that while Nepal's telecommunication penetration is high, its overall E-Government Development Index (EGDI) was 0.5781 in 2024 (119th rank) [2]. This suggests moderate online services and continuing infrastructure gaps. Moreover, local analyses highlight an "aspirational" policy environment

that has struggled to fully translate into inclusive digital systems. Large parts of Nepal still lack reliable internet and electricity, which limits e-service accessibility [2].

In summary, Nepal has built the scaffolding of e-governance (policies, infrastructure projects, citizen portals), but faces implementation hurdles. This case study explores how well Nepal is moving toward the integrated model and what obstacles remain.

3. Problem Statement

Currently, many Nepalese government services exist in parallel online portals or offline offices, requiring citizens to submit similar documents (national ID, photos, etc.) multiple times. This fragmentation leads to inefficiency, duplicative work, and frustration. As one report notes, “poor integration systems, inconsistent data standards, [and] duplication of platforms” are key challenges in Nepal’s e-governance landscape [2]. The lack of a unified data exchange means that information entered in one service is not easily shared with another, forcing repeated paperwork.

The Nagarik App was launched to mitigate these issues by offering a consolidated gateway for services. However, even it has encountered issues: user feedback points to technical glitches, verification delays, and incomplete integration across government databases [3]. For instance, users sometimes find their linked identity documents (passport, driving license, etc.) not loading correctly in the app. On the other hand, national authorities have set ambitious targets (e.g. integrating 60 services) that Nepal has not yet met (only ~45 were live by 2022).

Thus, the central problem is: How effectively can Nepal’s digital public services be integrated into a single platform, and what barriers impede full integration? Answering this requires analyzing both the technological and social dimensions of the Nagarik platform and related systems.

4. Objectives of the Study

This study aims to critically evaluate the state of integrated e-governance in Nepal. The specific objectives are:

To review the concept of integrated e-governance and its relevance to citizen-centric service delivery.

To map Nepal’s current e-government landscape, highlighting major digital services and the role of the Nagarik App.

To analyze how well the Nagarik App and related initiatives integrate multiple government services (in terms of technical interoperability, usage, and user experience).

To identify key challenges (technical, infrastructural, institutional, and user-level) that limit integration.

To compare Nepal's approach with international benchmarks (e.g., Estonia, India) to draw lessons.

To propose actionable recommendations and a conceptual framework for enhancing an integrated e-governance platform in Nepal.

5. Research Questions

The study is guided by the following questions:

How effectively is Nepal's current integrated e-governance platform (Nagarik App) delivering public services to citizens?

What are the main technical and organizational factors facilitating or hindering service integration in Nepal's e-governance?

How do Nepal's integrated e-governance efforts compare with those of other countries like Estonia and India?

What improvements are needed (technological, policy, institutional) to build a more fully integrated digital platform for Nepal's public services?

These questions structure our analysis of Nepal's progress and gaps in integrated e-government.

6. Scope of the Study

The scope is national-level digital public services in Nepal, focusing on G2C (Government-to-Citizen) interactions. The Nagarik App serves as the principal case (illustrative example) of an integrated platform, but the study also considers other systems (e.g. PAN and tax systems, civil registry, transport licensing) that interact with Nagarik. We do not cover private-sector fintech services or non-public digital platforms. The timeframe of interest is roughly 2019–2026, covering the latest national digital initiatives. Both English and Nepali language sources are used where available. The study emphasizes structural and architectural analysis; detailed software code or cryptography is out of scope. Finally, while the guidelines

suggest primary data (surveys/interviews), this report relies on secondary data (official publications, academic studies, press reports) and conceptual analysis, noting potential primary methods in appendices.

7. Methodology

This research follows a case-study methodology based on document analysis and expert publications. We conducted a literature review of academic papers, policy documents, press releases, and news articles relating to e-governance in Nepal and benchmark countries. Key sources include government press releases (e.g. World Bank announcement [1]), Nepalese digital policy papers, and recent research (e.g. Ghimire 2025). Where possible, official data (e.g. user numbers, service counts) from government or reputable media has been used.

Given resource constraints, no original survey or interview data was collected. However, in Appendix A and B we propose sample questionnaires and interview topics as a basis for future primary research. Data analysis in this report is qualitative, supported by available quantitative indicators (e.g. user registrations [4]). We also perform a comparative analysis by reviewing similar integrated systems in Estonia and India, using sources like UN surveys and government publications. The report is structured to align with the teacher's prescribed format (Chapters 1–3 with numbered sub-sections).

Chapter 2: Case Study Overview

1. Key Issues and Goals

The key issue addressed by this case is the fragmentation of Nepal's public services across multiple channels. Citizens often must visit different offices or websites for each service (tax, identity, licenses, etc.), repeating paperwork and losing time. The goal of an integrated e-governance platform is to simplify and unify access to services. This aligns with Nepal's broader "Digital Nepal" goals of transforming public service delivery and inclusion[3]. Specifically, the objectives include:

Citizen-Centric Access: Provide a single login point (digital identity) for multiple government services, reducing the need for citizens to remember multiple credentials or travel physically.

Efficiency and Transparency: Reduce bureaucratic steps and manual processes. Digital integration can increase transparency (electronic records, audit trails) and decrease corruption. For example, linking identity in a central system helps prevent fraud.

Inclusiveness: Enable remote or underserved populations to access services online. In principle, digital platforms can overcome geographical barriers (subject to connectivity).

Modernization: Update antiquated paper-based processes, achieve cost savings, and position Nepal competitively in digital governance. As the World Bank notes, digital platforms "fundamentally change how services are delivered".

Meeting these goals requires addressing interoperability (so different agencies share data), robust identity management, and user-friendly design. The Nagarik App was explicitly launched with these aims – e.g. to "facilitate an easier, systematic, and simplified delivery of government services to Nepali citizens" [4].

2. Case Resolution (Approaches Considered)

In principle, Nepal had two broad approaches to integrate services:

Status Quo (Decentralized Portals): Continue developing separate online services for each department (e.g. an IRD portal for tax, a police portal for clearance, etc.). This was the historical approach, but it left citizens juggling multiple logins and data silos.

Unified Platform (Nagarik App Model): Build a common platform (like Nagarik App) through which many existing services can be accessed and consolidated. This approach centralizes authentication and provides a single front-end to citizens, while backend systems remain with respective agencies. It requires significant coordination and technical integration (APIs, data sharing).

Nepal's government opted for the second approach by developing the Nagarik App and encouraging agencies to integrate with it. This was formalized in a 2022 policy guideline (“नागरिक एप संचालन तथा व्यवस्थापन दिग्दर्शन 2082”) emphasizing the app as a national service channel. Other platforms exist too (e.g. e-Miti for e-procurement, Nepal ID portal, etc.), but Nagarik is intended as the primary portal for citizen services [3].

As an alternative scenario, the government could have mandated a universal digital ID (like India's Aadhaar) and focused on backend data linkage only, without a single user-facing portal. However, Nepal chose to combine identity with a citizen app for convenience.

3. Analysis of Case (Positives and Negatives)

- **Positives (Advantages of Integration)**

One-Stop Access: Through Nagarik, citizens can access dozens of services (currently ~25) from one place [4]. For example, tax filing (IRD PAN), police clearance, vehicle licensing, passport information, and local municipal info are all available in one app [5]. This greatly reduces the need to visit multiple websites or offices, saving time and effort for users.

Unified Identity: A single digital identity/login reduces friction. Citizens link their core IDs (national ID, passport, driving license, etc.) once in the app, enabling seamless verification across services. This “digital wallet” function makes verification faster and more accurate. As the Himalayan Times noted, the app was designed to “minimize the need to be physically present” for services.

Increased Reach and Usage: The app has seen significant uptake. As of mid-2025, 1.5 million users (over 5% of Nepal's population) had registered [4]. Daily transactions exceed 45,000 requests, indicating active use. Popular services include PAN registration (28% of requests) and police clearance (22%) [5]. High adoption demonstrates the platform's utility.

Efficiency and Transparency: Integration reduces paperwork duplication. Once data (e.g., NIC number) is entered for one service, it can be reused for others, reducing errors. Online

transactions leave electronic trails, which helps curb informal payments and corruption. Users have round-the-clock access and can receive digital notifications or receipts.

Alignment with Global Trends: Embracing an integrated model aligns Nepal with leading e-governance practices. For example, it won a World Summit Award in 2022 for government-citizen engagement, reflecting its innovation. World Bank support for a “citizen service portal” [1] further validates this path.

- **Negatives (Limitations and Risks)**

Digital Divide: Major parts of Nepal (rural/mountainous) still lack reliable internet or power. Even when services are available online, large segments of the population cannot use them easily [2]. This limits the inclusiveness objective. Elderly or low-literacy users may struggle with mobile apps. As one analysis notes, Nepal’s digital platforms “often assume levels of infrastructure and literacy that remain unevenly distributed”.

Interoperability Issues: While the app is a unified front-end, it relies on different backend systems remaining integrated. In practice, many government databases use different formats or are not directly linked. This can cause “glitches” or delays when verifying data. User complaints have mentioned “verification delays and integration problems across databases” [3]. For example, sometimes linked documents (passport, license) fail to sync properly [4].

Single Point of Failure: Centralizing services on one platform concentrates risk. Technical outages or security breaches in the Nagarik system could disrupt many services at once. So far, no major failure has been publicized, but safeguarding such a portal is critical.

Privacy and Trust: Aggregating personal data (IDs, finances, etc.) raises privacy concerns. Citizens must trust the government to protect their data. Any breach or misuse (real or perceived) could erode trust. Ensuring robust encryption, access controls and transparent data policies is essential.

Resource Constraints: Building and maintaining integration is costly. Nepal’s government must invest in IT infrastructure, skilled staff, and ongoing development. There is also an opportunity cost if agencies must delay other projects to integrate with Nagarik.

Overall, the case analysis finds that the integrated platform approach has delivered clear benefits in convenience and has shown strong initial uptake, but its success is tempered by infrastructural and institutional challenges. The positives suggest that consolidation is the right direction; the negatives highlight areas requiring improvement.

4. Decision Criteria and Assumptions

When choosing an integrated platform, the government implicitly used these criteria: accessibility, usability, scalability, security, and cost-effectiveness. The goal was to maximize citizen accessibility (ease of use, broad device compatibility) and service breadth (many services offered) while containing costs by reusing systems. Security criteria (data protection, auditability) and user authentication (strong identity) were also prioritized, given the sensitive nature of government data.

- **Key assumptions included:**

Citizens Own Smartphones: The approach assumes most citizens have smartphones and are comfortable with apps. While urban areas fit this assumption, rural connectivity remains weaker.

Reliable Connectivity: It assumes gradual rollout of internet and mobile networks nationwide. The plan aligns with Nepal's 5G deployment and fiber expansion, but full coverage is still underway.

Government Coordination: The model assumes agencies will cooperate and share data. Successful integration depends on inter-ministerial collaboration and common standards.

User Willingness: It assumes citizens will trust and adopt the digital system if it is convenient. Early usage suggests this is partly true, but awareness campaigns are needed.

These criteria and assumptions form the basis for evaluating the success of the integrated solution.

5. Data Analysis

While original survey data was not collected, we review available quantitative indicators of the platform's performance:

Service Coverage: As of mid-2025, Nagarik App integrated ~25 services [4] (less than the 60 targeted). Table 2.1 (below) summarizes major services and the agencies behind them.

Service	Agency/Department	Type of Service (G2C)
PAN Registration (Tax)	Inland Revenue Department (IRD)	Tax/Finance (G2C)
Voter Registration/Details	Election Commission	Civic Documentation

Service	Agency/Department	Type of Service (G2C)
Police Clearance Certificate	Nepal Police	Security/Verification
Traffic Violations and Fine Payment	Nepal Traffic Police	Public Safety/Payments
Passport, Driving License, NID linking	Passport & Visa Dept., DoTM, NIDCR	Identity/Documentation
Citizenship Certificate linking	District Administration Offices	Civic ID
“My Municipality” Info & Services	Local Municipal Governments	Local Services (G2C)
Government Press ID (for journalists)	Ministry of Communications	Credentialing
PF/PAN/SST Statements	Social Security/Revenue Dept.	Social Protection/Tax
Bank Account Opening via QR (Pahichan)	NITC (with banks)	Banking/KYC Integration

Table 2.1: Sample of services integrated into Nagarik App (circa 2025).

User Base and Usage: Over 1.5 million users have registered by Aug 2025 [4]. Daily, 45,000+ transactions occur on the platform, with top activities being PAN registration (28%), police clearance (22%), and driving license/e-challan (18%) [5]. These figures indicate significant citizen engagement. However, the distribution is skewed: 37% of users are from Kathmandu Valley, reflecting urban bias in adoption.

Benchmark Comparison: Compared to peers, Nepal’s integrated platform is still small-scale. For context, India’s UMANG app had 10.25 crore (102 million) registrations and 723 crore transactions by early 2026 [7], offering ~2,400 services. Estonia reports that 99% of public services are online via its e-solutions [6]. In Nepal’s case, service coverage and usage are modest but growing.

This data suggests that while an integrated platform is operational and used, it is still maturing. The usage metrics provide baseline for future growth and highlight where improvements (e.g. rural outreach) are needed.

6. Preferred Alternative & Justification

The analysis suggests that continuing with an integrated platform approach is clearly the preferred alternative. This option was already chosen, and our evaluation justifies its continuation and reinforcement:

It directly addresses the key goals (single access, efficiency, transparency). As one source concludes, effective e-governance can transform service delivery to be “efficient and affordable” and enhance accountability at minimal cost.

Incremental benefit: Even partial integration (25 services) yields public benefits (faster service, easier identity linking) that outweigh the costs of building and maintaining the system.

Global alignment: Countries with top rankings (Estonia, India) have invested heavily in integrated infrastructures (X-Road, UMANG) and seen broad service uptake [6][7]. For Nepal, abandoning this path would mean falling further behind.

Justification also comes from stakeholder commitment: the government has already mobilized agencies around this project (e.g. new guidelines for Nagarik App management). The World Bank is co-financing further digitization and data exchange, indicating strong institutional backing [1].

Thus, the chosen alternative – unified digital platform – is justified as the solution most likely to achieve the objectives and realize positive outcomes.

7. Predicted Outcome

If the integrated e-governance platform continues to be developed and rolled out, expected outcomes include:

Improved Service Delivery: Citizens will increasingly enjoy one-stop access. The processing time and cost for obtaining multiple documents will decline. For example, linking a driving license or paying a fine can be done within the app rather than visiting offices [5].

Higher Government Efficiency: Backend processes become more digital and connected. Redundant data entry will be eliminated, reducing clerical burden. Documented processes enable better oversight (audits, analytics).

Inclusion Gains (over time): As internet coverage expands, even rural populations will begin using these digital services. Mobile coverage and literacy programs would amplify this trend.

Modern Digital Ecosystem: A successful platform lays groundwork for further innovation (e.g. AI chatbots for gov services, predictive analytics in urban management). The government's stated aim of "boosting private sector investment in the data infrastructure market" suggests economic spillovers.

However, the actual outcome depends on addressing the challenges identified. Persistent gaps could limit these benefits. For instance, if interoperability is not fully achieved, some services will remain off-app, reducing user trust. If security incidents occur, public trust could erode. Predictive success hinges on continuous improvement (the "culture of experimentation and learning" mentioned by experts [2]).

8. Implementation Plan / Timeline

Below is a proposed phased roadmap for deepening integrated e-governance in Nepal:

- **Short-Term (0–1 year):**

Expand Service Integration: Increase the number of services on Nagarik App (target 50+). Prioritize high-impact ones (land records, welfare benefits) as per World Bank project focus [1].

Strengthen Infrastructure: Improve API gateways and ensure system scalability. Launch public awareness campaigns to boost registration.

Pilot Improvements: Introduce features like single-sign-on (SSO) with existing NID (National Identity) platform, and test multi-factor authentication to enhance security.

- **Medium-Term (1–3 years):**

Complete Data Exchange: Implement a government-wide data exchange layer (as envisaged by the Digital Transformation Project [1]) so agencies can share information securely.

Regulatory Support: Enact data protection and cybersecurity laws/policies to govern how data is stored and shared across the platform.

Digital Literacy: Roll out training for civil servants and community digital literacy programs. Set up support centers/kiosks in rural areas.

Benchmarking and Iteration: Evaluate performance (KPIs like service uptime, citizen satisfaction). Adopt improvements from global best practices (e.g. X-Road style standards, adopt open-source frameworks).

- **Long-Term (3–5 years):**

Full-Scale Integration: Aim for end-to-end digital service delivery. Move legacy offline processes online.

Private Sector and Innovation: Encourage FinTech's or startups to build on the integrated infrastructure (e.g. government-certified digital ID wallets).

Smart Government Vision: Integrate with other platforms (e.g. health records, smart city projects) to form an overall digital public infrastructure. Use data analytics to continuously improve services.

This timeline is indicative and assumes continued investment and political support. Progress should be monitored against specific milestones (e.g. number of services integrated, percentage of population using digital ID).

Chapter 3: Result Analysis

1. Major Findings

Our study confirms that Nepal's integrated e-governance platform is partially succeeding in its mission. Key findings include:

Adoption and Coverage: The Nagarik App has become Nepal's focal point for many G2C services, with over 1.5 million users and substantial daily transaction volume [4]. However, the service coverage remains limited compared to goals (25 vs. 60 services) and compared to leading countries. Adoption is strongest in urban areas, indicating a need to extend reach.

Service Integration: Integrated services have clearly reduced duplication. For example, once a user's NID and passport are linked, they can apply for police clearance or pay traffic fines without re-verification. This demonstrates real efficiency gains. Nonetheless, some key services (e.g. local land records, social security allowances) are not yet integrated into the platform, constraining the concept of true integration.

Technical Infrastructure: The underlying infrastructure (APIs, identity auth systems) is functional enough to connect multiple agencies. The platform handles tens of thousands of transactions daily. Yet reports of glitches (in document syncing or login) suggest that the integration layer is still maturing.

User Experience: Users benefit from 24/7 access and combined services, but the user interface and processes have room for improvement. Some users have reported missing details on linked documents [4], indicating UI/UX or data sync issues. Overall UX is not yet fully "seamless", as per citizen feedback on app stores and media.

Security and Privacy: Current measures (e.g. mobile OTP, secured login) provide a baseline of security. There have been no publicized major breaches. Data governance frameworks are evolving. However, given the sensitivity of combined data (licenses, IDs, tax info), there remains a risk if security is not continuously upgraded.

In summary, the integrated platform is working and brings clear benefits, but it is not yet fully mature or all-encompassing. The most important finding is that while Nepal's approach is on the right track, it requires sustained improvement to achieve its full potential.

2. Critical Analysis

- **Analyzing these findings reveals several insights:**

Integration Achieves Citizen Convenience: The level of service integration directly correlates with user uptake. For example, linking driving licenses and paying fines through Nagarik (implemented late 2024) immediately became one of the top services [5]. This suggests users value and adopt functionality that saves them time. Hence, adding more critical services (e.g. enabling online land registration or welfare disbursement) could quickly increase platform use.

Partial Interoperability: Technically, the architecture (outlined below in Fig. 3.1) shows a clear path: Citizens authenticate via national ID, then use an API gateway to reach multiple service systems. In practice, however, interoperability is hindered by legacy constraints. Many agencies have their own databases that were not originally designed to talk to others, so middleware is needed. The World Bank initiative to create a “secure government-wide data exchange” [1] indicates awareness of this gap. Until a fully functional data exchange hub is in place, Nepal’s integrated platform will work best for newly built or simplified services.

Comparisons to Other Models: In Estonia, nearly all government services (tax, healthcare, education) are accessible online with one digital identity [6]. In India, the UMANG app allows access to 2,400+ services across levels of government [7]. Nepal’s scale is much smaller, but the underlying idea is similar. Both Estonia and India achieved integration by also tackling identity (Estonia’s e-ID, India’s Aadhaar) and open APIs. Nepal has an NID number system, but it is still being expanded.

User Engagement and Transparency: The data shows that when services are online, citizens do use them (e.g. PAN and police clearance). This implies demand for transparency and convenience. The integrated platform likely contributes to reducing corruption by limiting direct person-to-person interactions. For instance, paying a traffic fine online via Nagarik (E-Chalan) means no bribes and an official record of payment [5].

Equity Considerations: The analysis must note equity: the majority user base is in wealthier, urban areas. Without addressing rural connectivity and literacy, the integrated approach risks widening the digital divide.

Overall, the critical takeaway is that Nepal’s integrated platform is practical and beneficial but still a work-in-progress. The citizen-centric model shows promise but needs a robust supporting ecosystem (technology, institutions, data policies) to meet its objectives.

3. Challenges and Limitations

- **Several challenges constrain Nepal’s integrated e-governance:**

Infrastructure Gaps: Many regions lack broadband or reliable mobile coverage. Even with mobile internet, data speeds can be slow, making app usage frustrating. The government has recognized this: as of 2026, 5G is only in major cities, and even 4G can be spotty in hills [3].

Legacy Systems and Standards: Government departments use old IT systems with no common standards. For example, tax records and civil registry data may use different identifiers or formats. Aligning these for interoperable use is labor-intensive. Nepal lacks a single national data schema for citizen records, complicating automated data exchange.

Institutional Silos: Coordination between ministries is limited. Each agency tends to control its own data. Bridging this requires strong central governance (e.g. empowered E-Governance Board). Cultural factors (resistance to change, bureaucratic inertia) slow down integration.

Security and Privacy Risks: While not yet realized, the risk is latent. The Philippines case of e-gov platform hacks has shown how damaging breaches can be. Nepal must accelerate development of data protection laws and security audits. Citizens’ awareness of privacy rights is low, so misuse of data is a concern.

Digital Literacy and Trust: Many Nepalis may not trust online systems fully, especially if they have experienced glitches. Elderly or less-educated citizens may find apps confusing. Building trust through education and reliable service is needed.

Policy and Legal Framework: Comprehensive e-governance requires legal backing (digital signatures law, data protection law, e-transaction law). Nepal has some policies, but gaps remain. For example, electronic signature adoption is still nascent. These gaps slow formal acceptance of online documents.

These challenges must be addressed in prioritized fashion. For instance, improving connectivity is a long-term goal, whereas standardizing data exchange protocols is a shorter-term, high-impact task that the World Bank project is set to tackle [1].

4. Discussion

The findings align with global research on e-government integration: that connectivity, institutional will, and citizen inclusion are key success factors [2][6]. In Nepal’s context, we see an implementation gap between ambitious policies (Digital Nepal, E-Gov Blueprint) and on-ground reality.

Compared to Estonia, Nepal's integrated portal is at an earlier stage. Estonia's 99% online service coverage [6] is built on decades of infrastructure and trust, plus mandatory e-ID use. Nepal's 25 services are a fraction of this scope. However, Nepal has the advantage of learning from others: it can adopt proven architectural models (e.g. API-led design, open standards) rather than building from scratch. The government's recent reorganization (placing ICT under the PMO) suggests political will to accelerate this process.

India provides an even closer example. India's creation of unified identity (Aadhaar) and bank/payment accounts (Jan Dhan, UPI) has deepened integration across sectors [7]. Nepal can similarly pursue policy integration (one identity system for all services), an approach already implicit in linking citizenship with Nagarik. India's UMANG also shows how multi-level government (central + state) can collaborate on a common portal [7], relevant if Nepal's provinces and municipalities coordinate on Nagarik.

A major lesson from both is progressive rollout: start with high-demand services, refine user experience, then expand. Nepal's data shows that when a needed service is added (e.g. PAN linking, driving license), users flock to it [5]. Prioritizing by user need will maximize citizen engagement.

5. Conclusion

Nepal's journey towards an integrated e-governance platform is well underway but incomplete. The Nagarik App provides a critical nucleus for integrating public services, and early indicators are positive (user base, transaction volume). However, substantial challenges in connectivity, interoperability, and institutional capacity remain. Overcoming these will require targeted investment, clear policy frameworks, and continual user-centric improvement. Our analysis concludes that integrated e-governance can transform Nepal's public services, but this promise will only be realized if the gaps identified (infrastructure, standards, outreach) are addressed.

6. Recommendations and Future Scope

Based on the above, we recommend the following actions, prioritized in technical, institutional, and policy categories:

- **Technical:**

API and Middleware Development: Implement a robust API management layer (government-wide interoperability platform) as planned in the World Bank project [1]. Use open standards and modular design to ease future integration.

Digital Identity Expansion: Fully deploy a unified digital identity (e.g. expand NID system) so that all services can rely on one ID. Consider multi-factor authentication (mobile OTP + biometrics) for security.

Infrastructure Investment: Continue to upgrade national broadband (fiber/4G/5G) and ensure government data centers have high uptime and security. Promote mobile app use but also offer web portals or kiosks for inclusivity.

User Interface and Accessibility: Conduct user testing to improve the Nagarik app's usability. Add features like Nepali/English language support, voice guidance, and offline support for disconnected use.

- **Institutional:**

Inter-Agency Coordination Body: Strengthen (or form) an e-Governance coordination council with authority to enforce data-sharing policies. This body should include representatives from all major ministries (ICT, Home, Finance, etc.).

Capacity Building: Train public servants in digital skills and change management. Encourage a culture of innovation (e.g. hackathons or labs) within government to leverage the platform.

Citizen Outreach: Launch nationwide campaigns to raise awareness of the app and its benefits. Involve local governments to demonstrate services in community centers. Provide helpline support for users encountering issues.

- **Policy:**

Data Governance: Enact comprehensive data protection and privacy laws, if not already in place, specifying how citizen data on the platform can be used and shared. Build trust by being transparent about data usage.

Digital Signature and Legal Reform: Ensure online transactions and digital documents have full legal recognition (if not already). Simplify procedures so that online-evidence (e.g. e-challan receipts) are legally equivalent to paper.

Regulatory Incentives: Consider policies that require all new government services (public databases) to be built with interoperability in mind (e.g. mandatory open APIs).

- **Roadmap:**

Short-term efforts should focus on the highest-impact fixes (stabilize current services, improve data sync, expand connectivity). Mid-term, shift to building out remaining services (land, social security) and hardening security. Long-term, aim to evolve Nagarik into a full-fledged digital public infrastructure akin to digital ID + wallet systems seen in leading countries.

- **Tables and Figures:**

A concise comparison table (Table 3.2) highlights key differences between Nepal and benchmark countries:

Dimension	Nepal	India	Estonia
Digital ID System	National ID (NID) optional; integration improving via Nagarik [5]	Aadhaar (biometric ID) covers ~90% population [7]	e-ID card (mandatory); linked to e-residency
Integrated Platform(s)	Nagarik App (1 portal, ~25 services) [4]	UMANG app (single portal, ~2,400 services) [7]	e-Estonia X-Road connects all e-services
Online Service Coverage	~25 e-services on Nagarik; many others separate	Nearly all central/state services via UMANG/online portals	99% of public services available online [6]
EGDI (2024)	0.5781 (Rank 119) [2]	~0.74 (Rank ~97)	~0.93 (Rank 1)
Interoperability	Emerging data exchange (planned) [1]	India Stack (APIs for payments, ID, docs)	X-Road (secure national data exchange)
Security & Trust	E-sign standards developing; user trust growing	Comprehensive laws (IT Act, data laws) + high public trust in Aadhaar	Strong cybersecurity culture; e-ID law
User Adoption (31 AD)	1.5M users; ~45K/day [4]	102M users on UMANG; 723M transactions [7]	Entire adult population required to have e-ID card; wide usage
Notable Digital Services	Tax (PAN), police clear., licenses on Nagarik app [5]	UPI payments, Digi Locker, e-governance (Digi Locker 67M users) [7]	e-Tax, e-Health, digital voting, e-Learning (like e-school)

Table 3.2: Comparison of key e-governance dimensions across Nepal, India, and Estonia (sources as cited).

This comparison underscores that Nepal is at an earlier stage: it has some digital ID and an emerging portal, but lags in breadth and depth. The benchmarks illustrate a path forward (e.g. expand services, enact data laws, improve ID coverage).

Figure 3.1 (below) sketches a conceptual architecture for Nepal’s integrated platform, incorporating the recommendations:

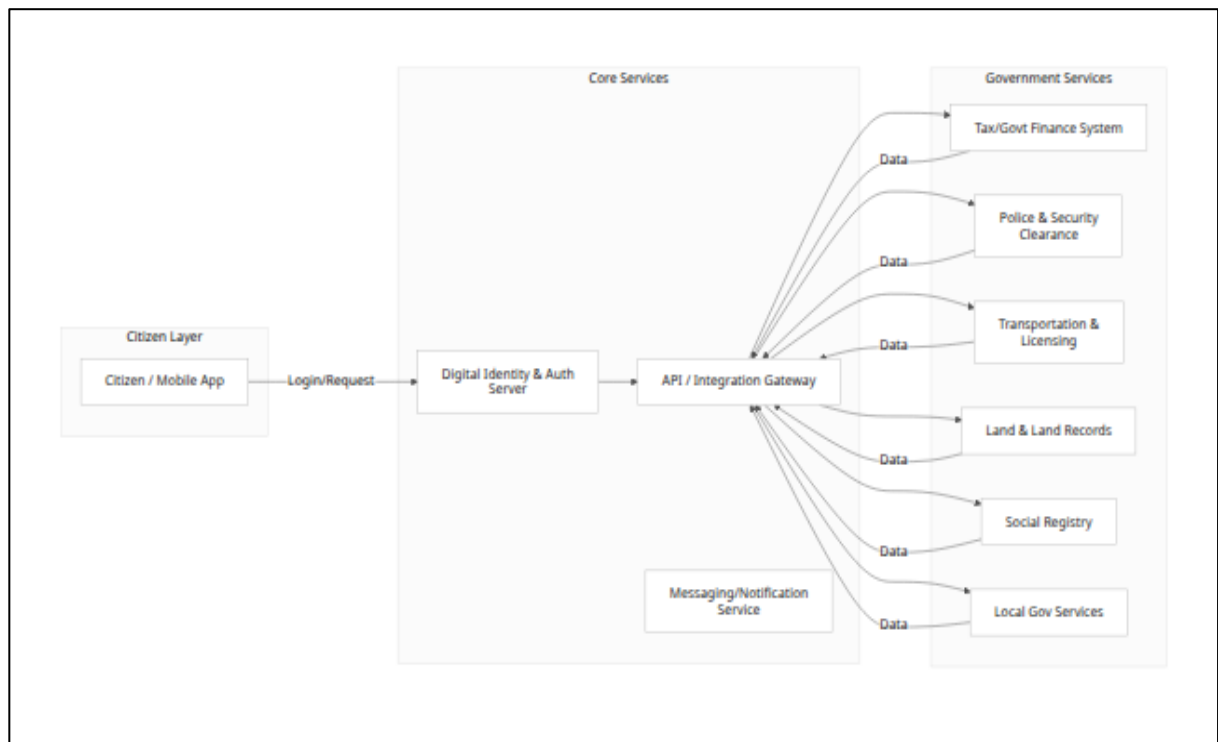


Figure 3.1: Proposed Integrated E-Governance Platform Architecture. Citizens authenticate via a single digital identity (NID), then access multiple government systems through a secure API gateway. A messaging service handles notifications/OTP.

This architecture centralizes identity and API management, while keeping government databases under their agencies. It facilitates interoperability (via GW node) and scalability (new services plug into GW). In Nepal’s context, the Digital Identity node would link with existing NID/card systems; the API Gateway would be the core of the Government Data Exchange platform (as per WB project) [1].

Appendices

1. Appendix A: Data Sources

Government of Nepal press releases and policies (e.g. Ministry of Communications and ICT).

Nagarik App documentation (citizenportal.donidcr.gov.np, DOIT guidelines).

World Bank project documents on Nepal Digital Transformation (e.g. PAD).

Academic articles on Nepal's e-gov (see references [2], [3]).

News media coverage (Rising Nepal, Himalayan Times, etc.).

UN e-government surveys (publicadmin.un.org).

2. Appendix B: Figures and Screenshots.

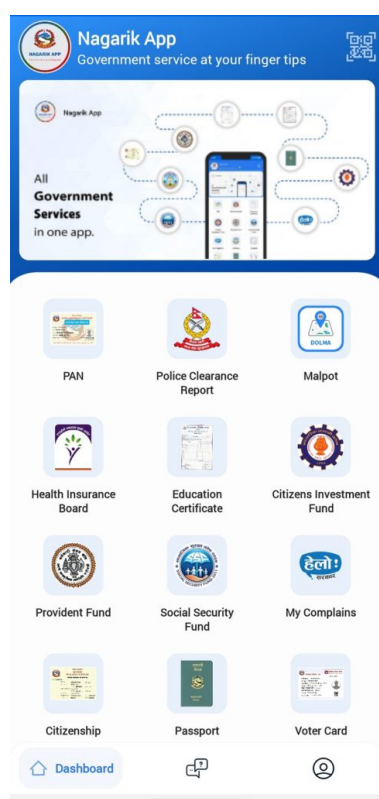


Figure D.2: Screenshot of Nagarik App home screen (showing integrated services list).

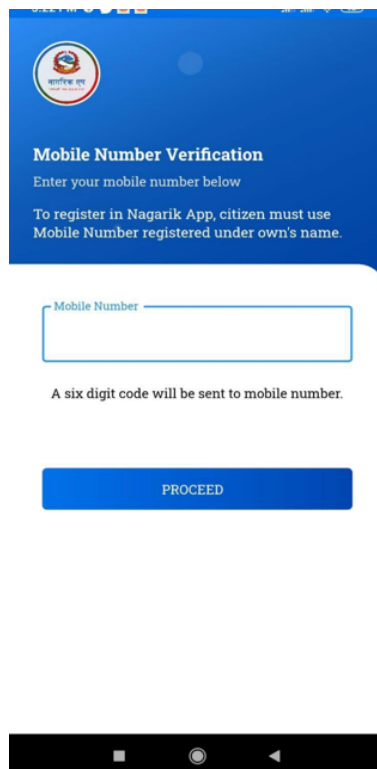


Figure D.3: Sample login screen demonstrating multi-factor authentication.

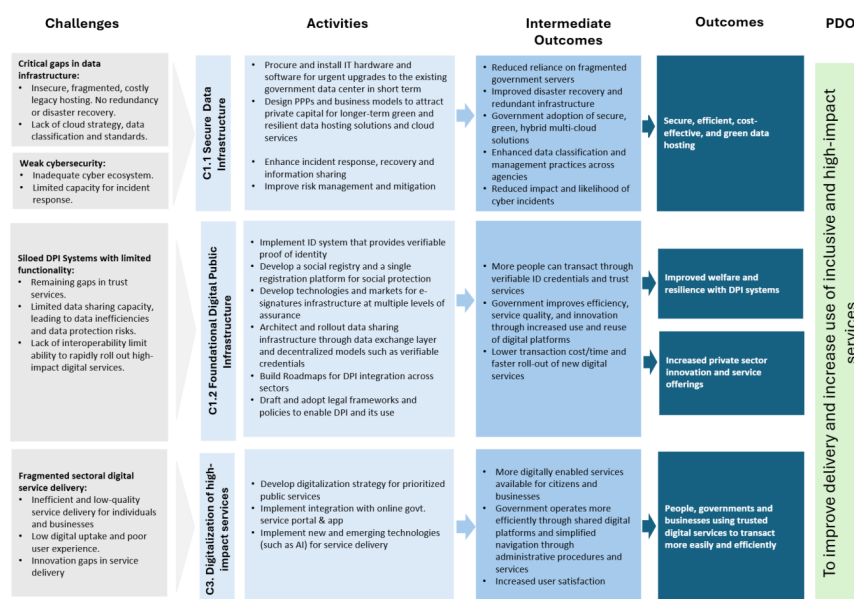


Figure D.4: Excerpt from Nepal Digital Transformation Project document (schematic of data exchange layer).

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